



Design of Model Predictive Control via Learning Automata for a Single UAV Load Transportation

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Agenda



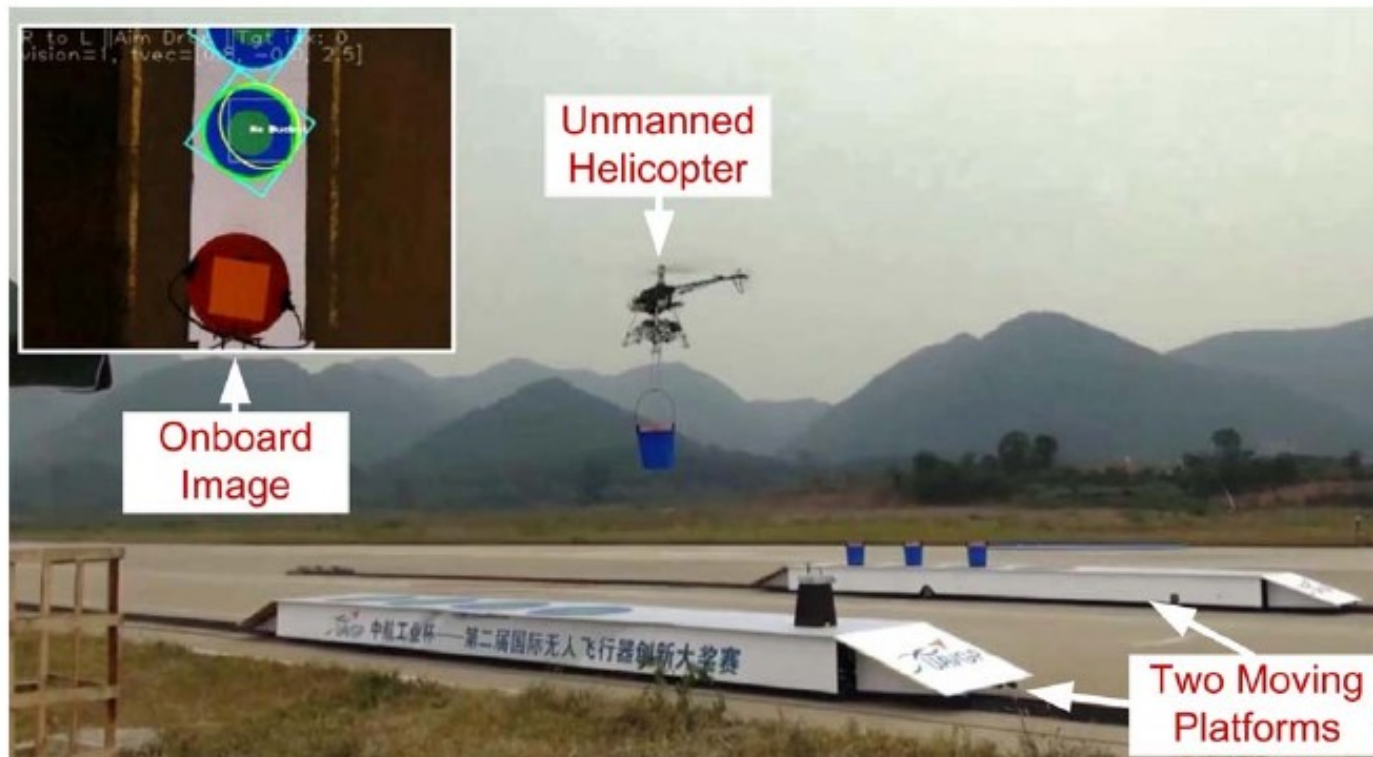
- Introduction
 - Load transportation
 - Controllers
- Problem Formulation
 - Simulation Environment
- Control System Design
 - NMPC
- Learning Control Algorithm
 - Learning process
- Results

Introduction



- UAVs
- Controllers
- Load Transportation
- Objective: “Use a Learn Automata to tune NMPC controllers for trajectory tracking during load transportation.”

Load Transportation



Zhao, 2015

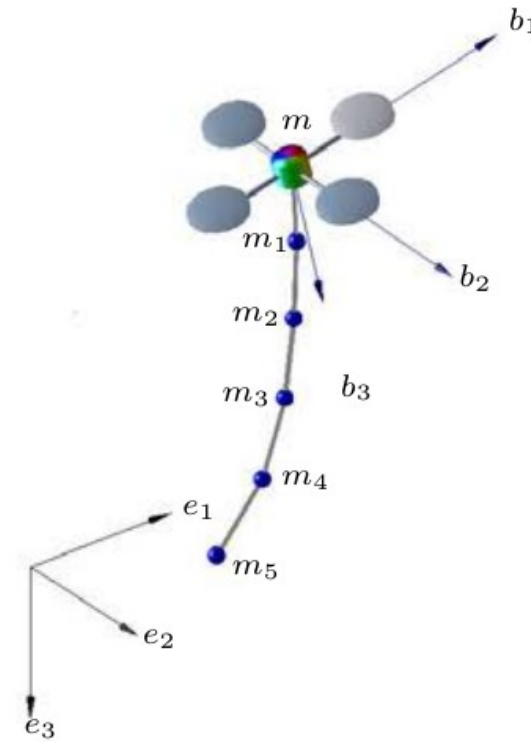
- 2013 International UAV Innovation Grand Prix;
- Ship-to-ship vertical replenishment;
- Real time vision system;

Controllers



Khalifa, 2013

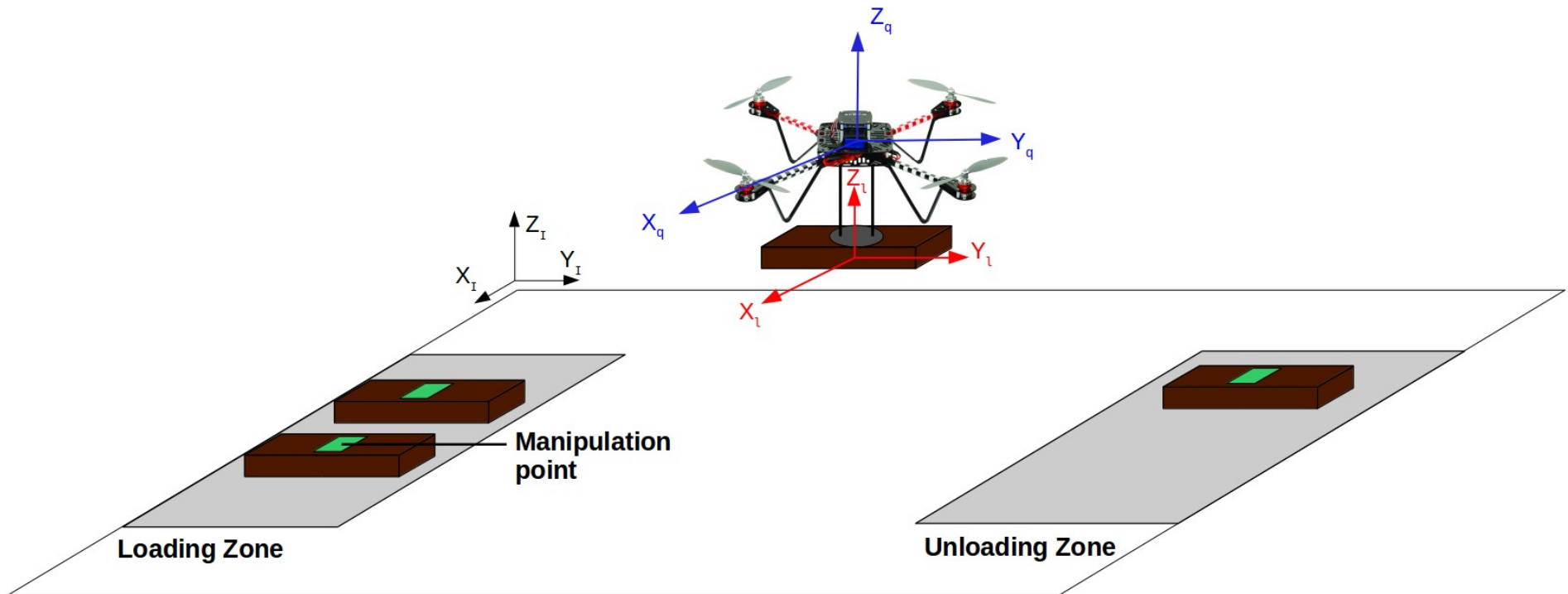
- Direct Fuzzy Logic Controller
- Fuzzy Model Reference Learning Control



Goodarzi, 2014

- Payload connected through a flexible cable; (mod. as serially-connected links);
- Geometric controllers (linearized, asymptotically);

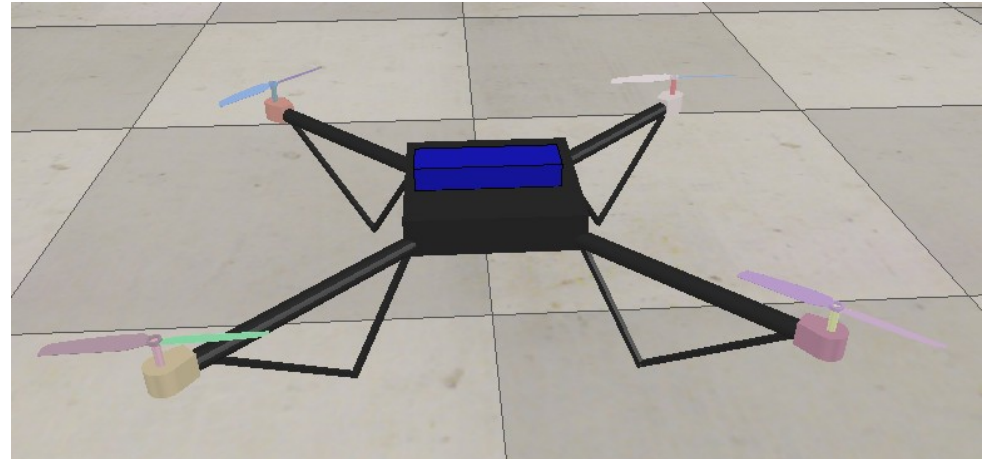
Problem Formulation



Simulation Environment



Parallax Elev8 Model

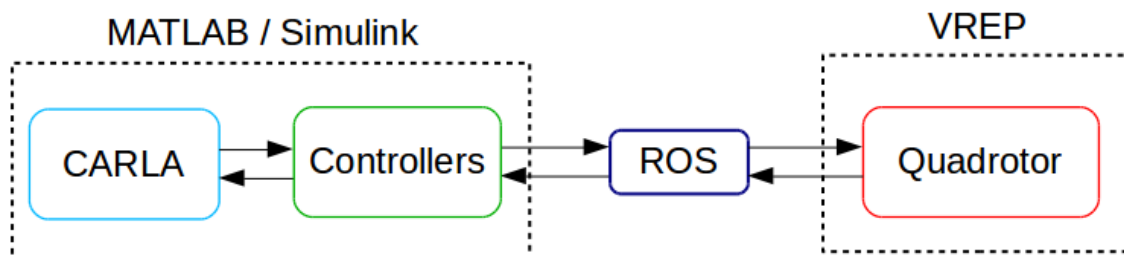
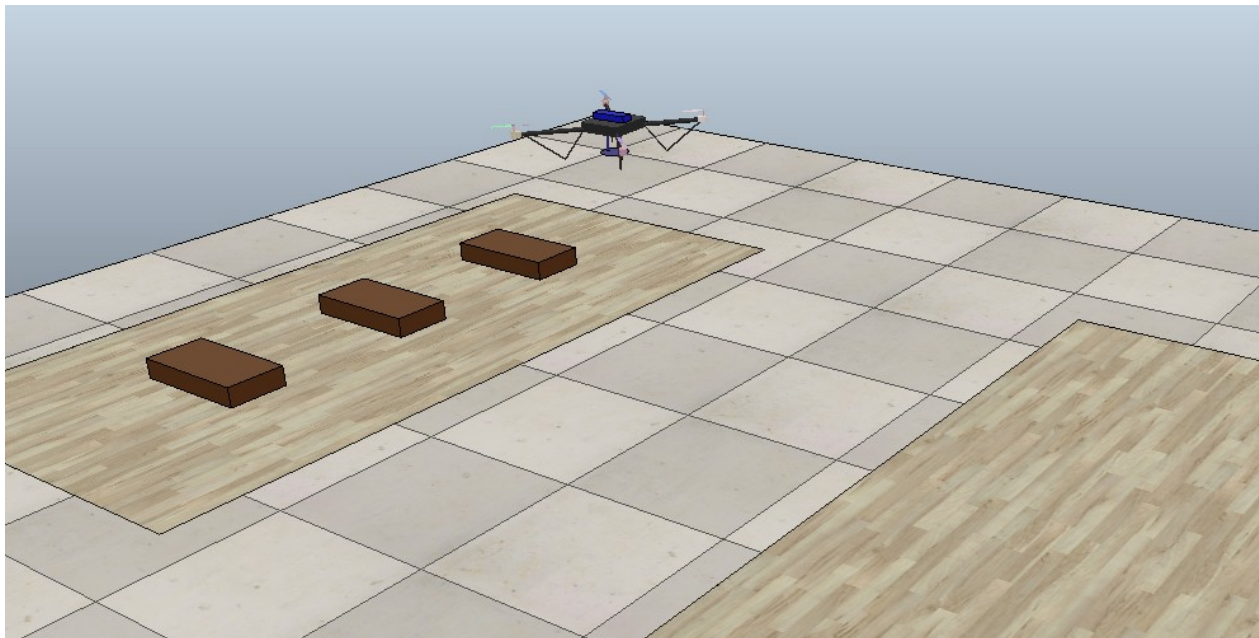


Simulated Model
used in V-REP

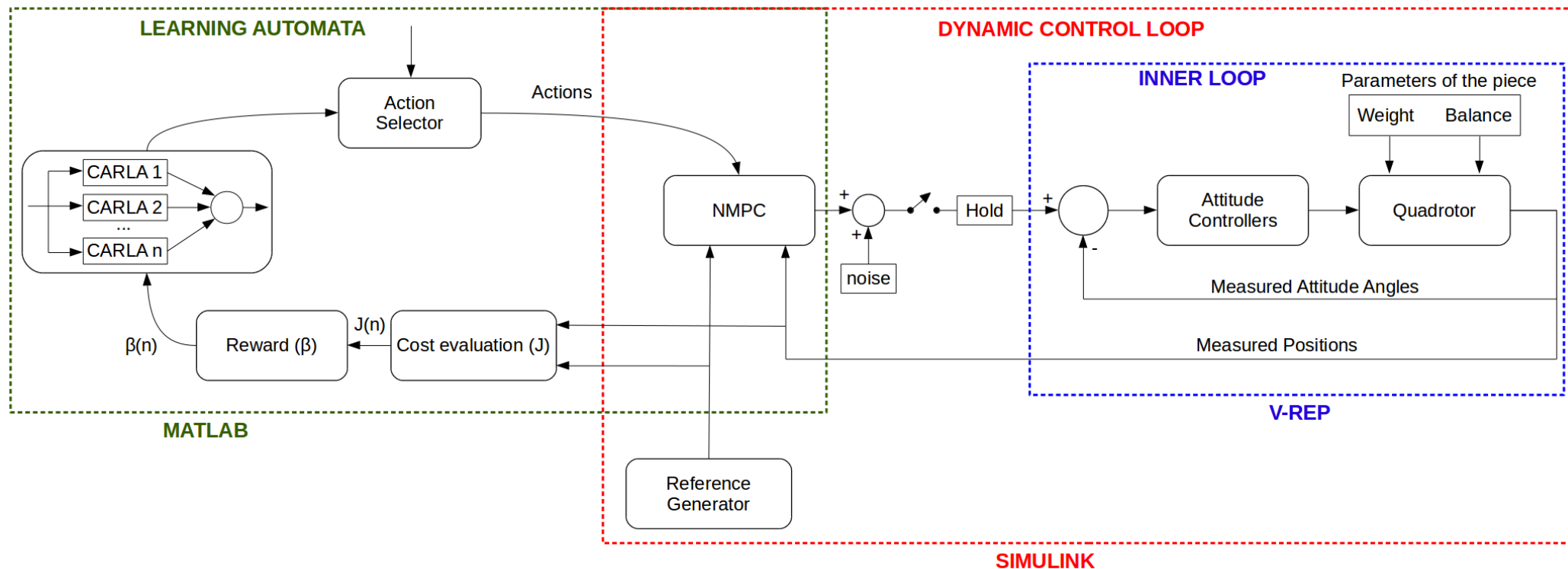
Simulation Environment



V-REP Simulation Software



Control System Design



NMPC

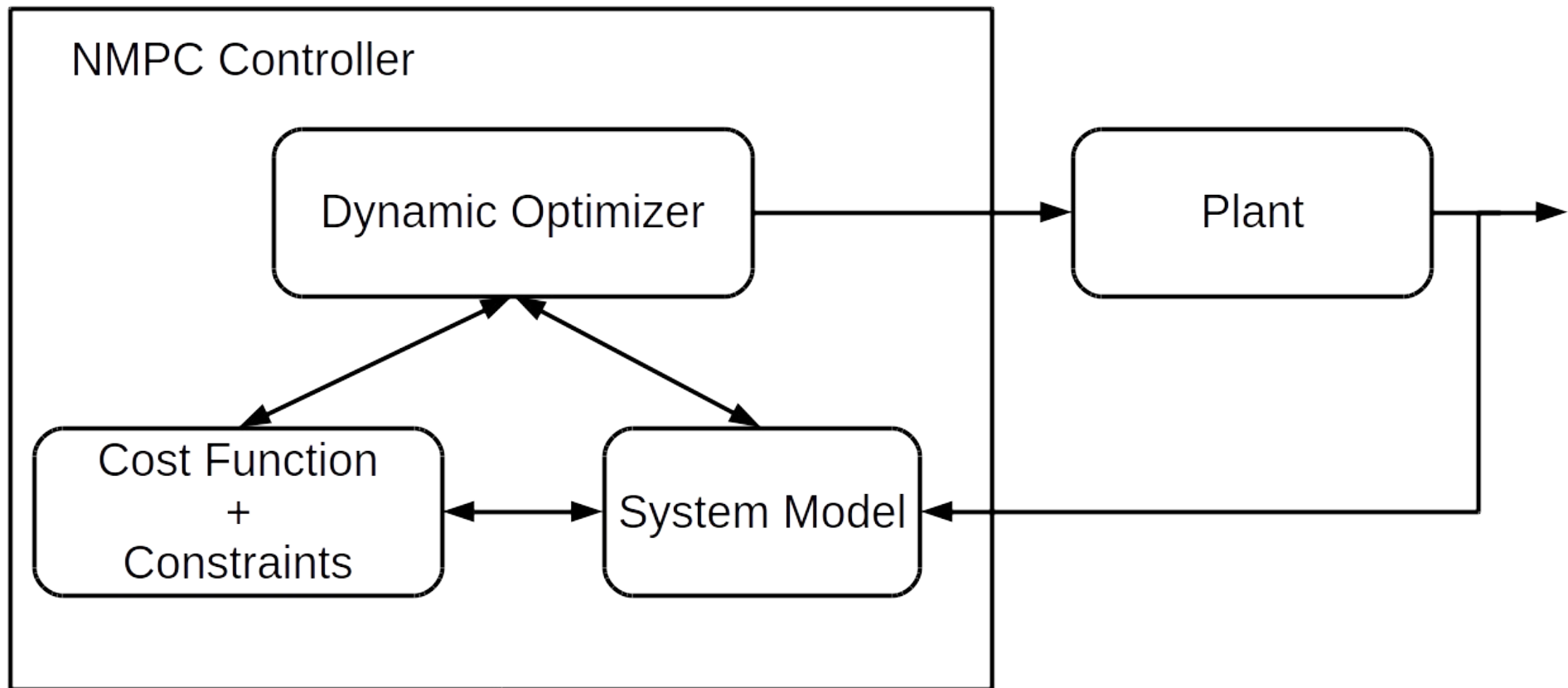


- Cost Function

$$\begin{aligned} J_{N_p}(\bar{x}, \bar{u}) &= \min \left(\bar{x}_{N_p} - \bar{x}_{N_p}^* \right)^T P \left(\bar{x}_{N_p} - \bar{x}_{N_p}^* \right) \\ &+ \sum_{k=0}^{N_p-1} \left(\bar{x}_k - \bar{x}_k^* \right)^T Q \left(\bar{x}_k - \bar{x}_k^* \right) \\ &+ \left(\bar{u}_k - \bar{u}_k^* \right)^T R \left(\bar{u}_k - \bar{u}_k^* \right) \end{aligned}$$

- Q and R are adjusted by the Learning Automata
- Optimization done by “quadprog”

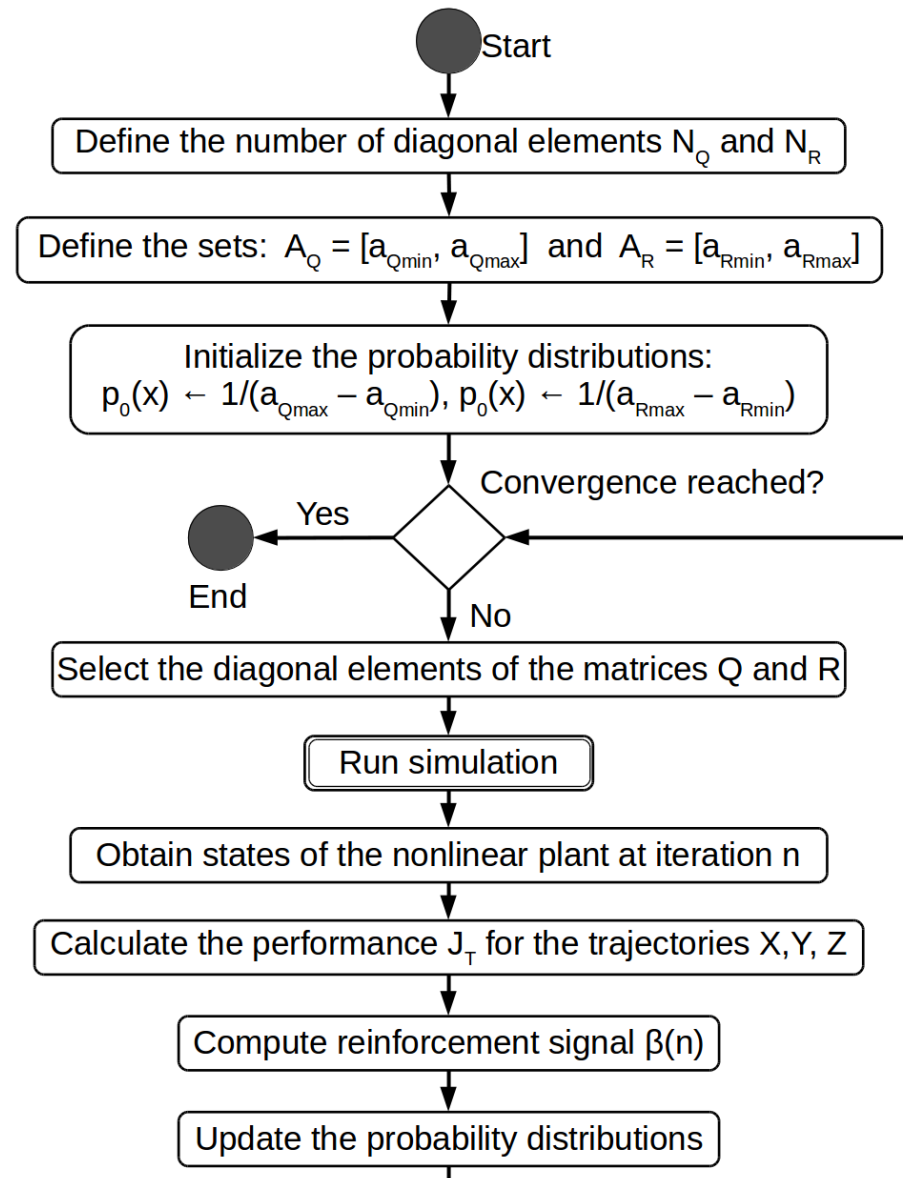
NMPC algorithm



Constraints:

- Limits for roll, pitch, yaw angle;
- Limits for thrust

Learning Automata Algorithm

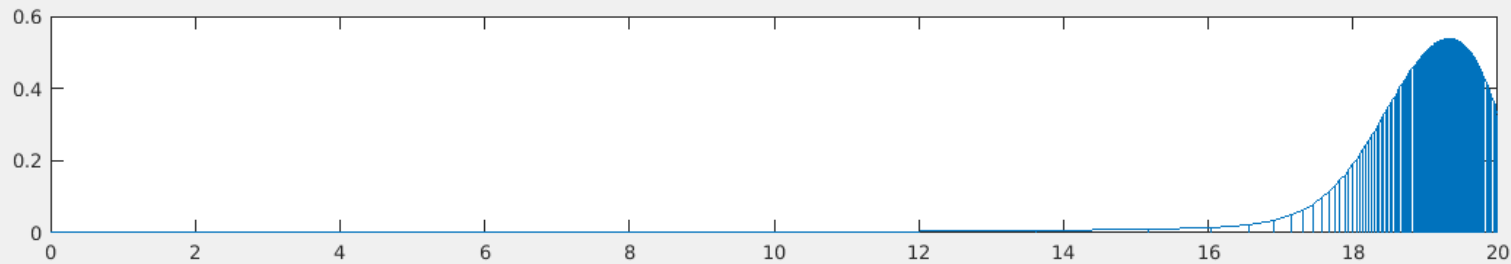


Learning process - CARLA

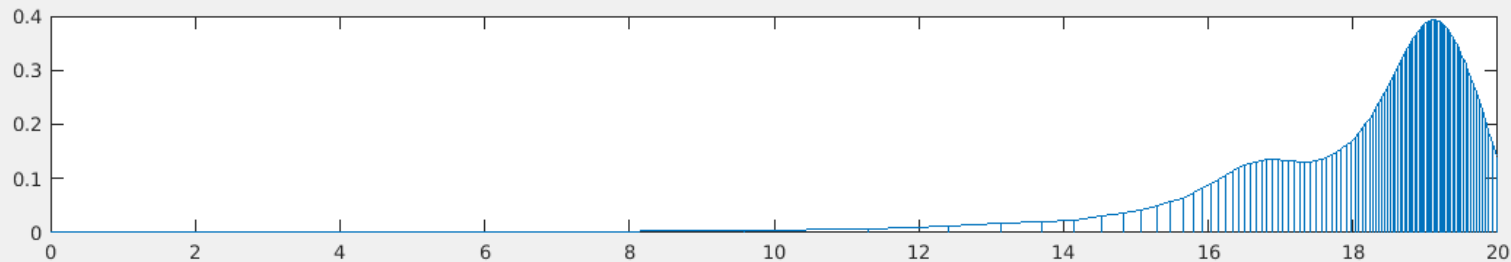


Actions

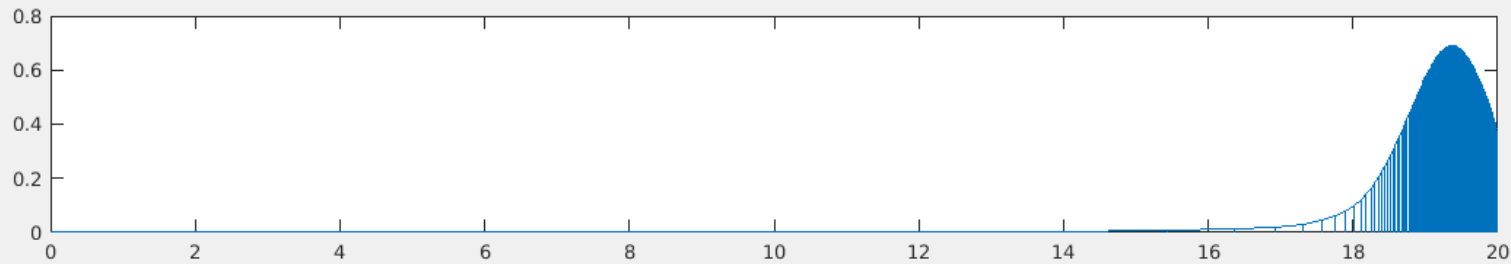
a1



a2



a3



Probability Distributions

Convergence: Max. probability
more than 80%

Learning Process



- Cost Function

$$J_T(n) = \int e^2 dt$$

Error:
pos. desired – actual pos.

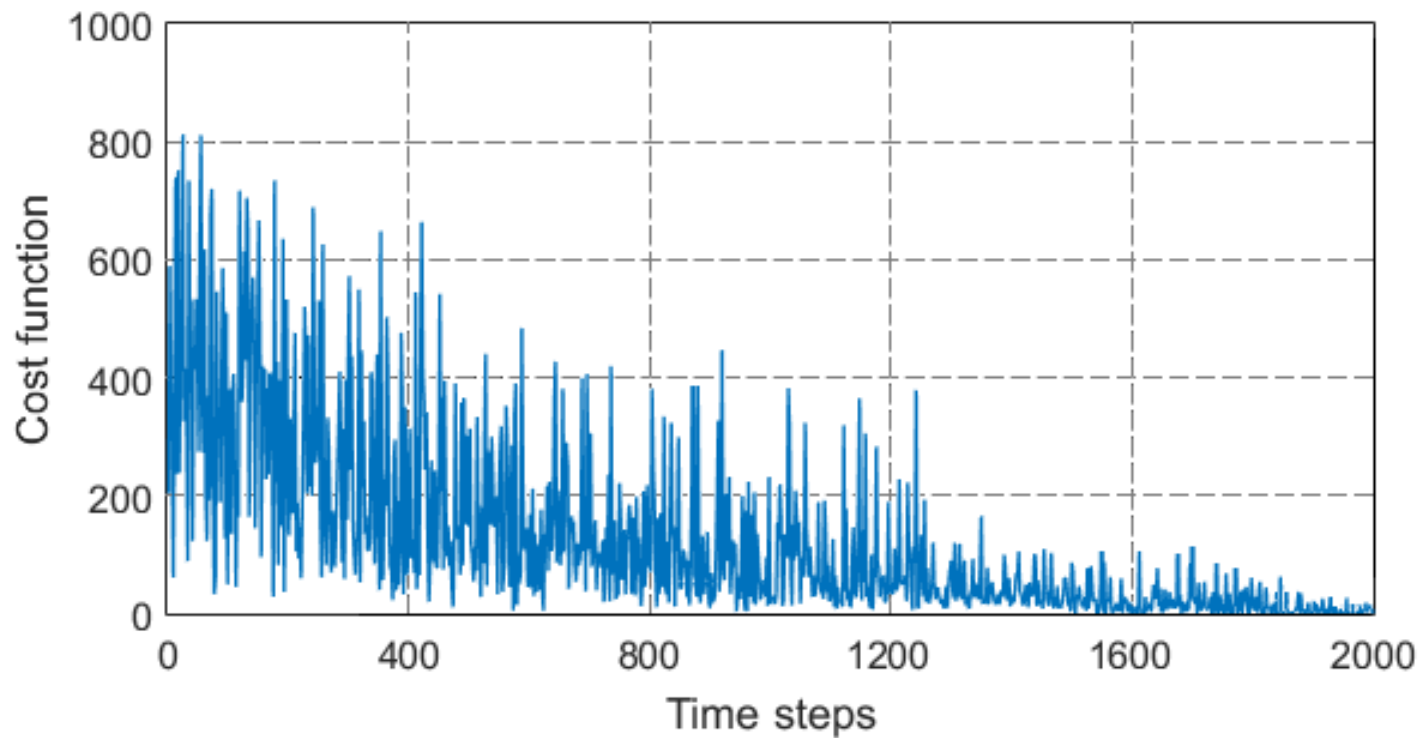
- Reward Function

$$\beta(n) = \min\left(\max\left(0, \frac{J_{T,med} - J_T(n)}{J_{T,med} - J_{T,min}}\right), 1\right)$$

Results



- Learning process



Results



- Evaluation Process

RMS AND CONTROL EFFORT WITHOUT LOAD

	RMS x(m)	RMS y(m)	RMS z(m)	Control Effort $\int u_{\phi} dt$
Classic PID	0.04530	0.09640	0.04330	0.00600
Nonlinear PID learning-based	0.03480	0.03960	0.01770	0.01580
NMPC	0.03412	0.03721	0.01534	0.00391

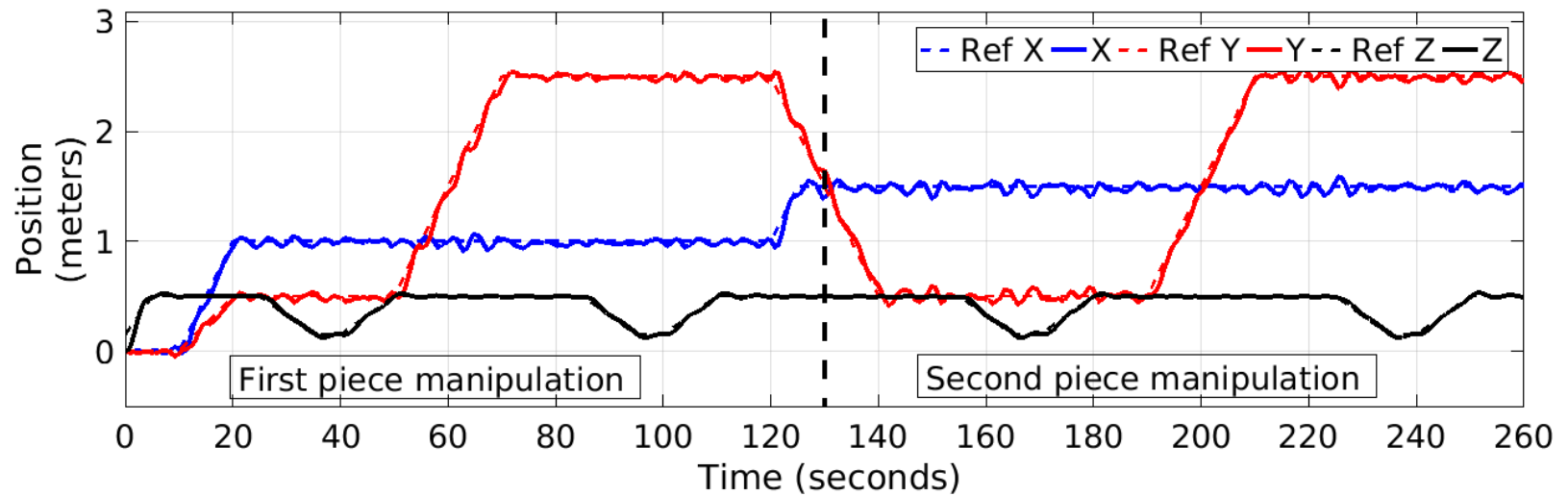
RMS AND CONTROL EFFORT WITH UNBALANCED LOAD

	RMS x(m)	RMS y(m)	RMS z(m)	Control Effort $\int u_{\phi} dt$
Classic PID	0.04520	0.09630	0.04350	0.00510
Nonlinear PID learning-based	0.03440	0.03980	0.01840	0.00970
NMPC	0.03021	0.03574	0.01484	0.00149

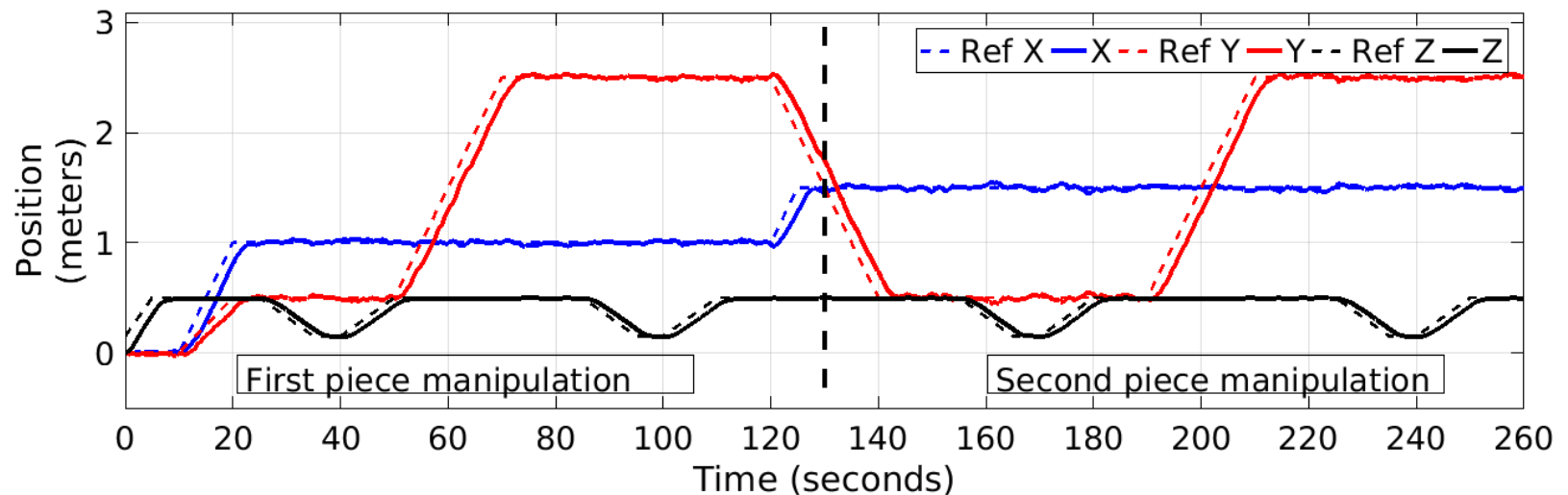
Results



Nonlinear PID
learning-based



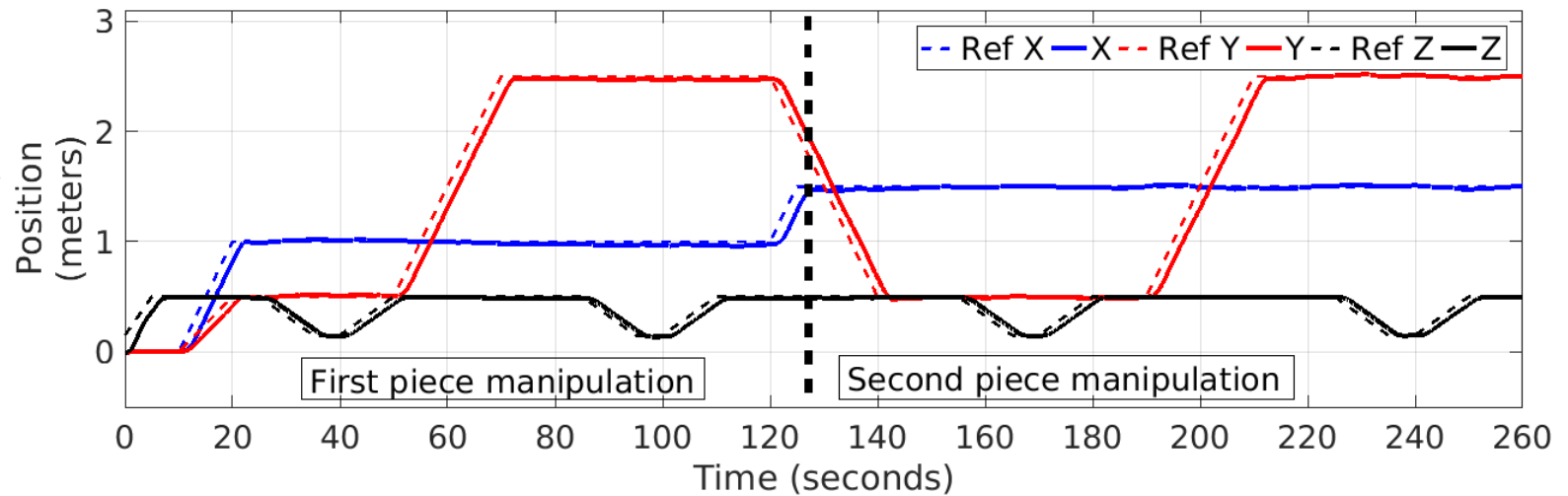
Classic PID



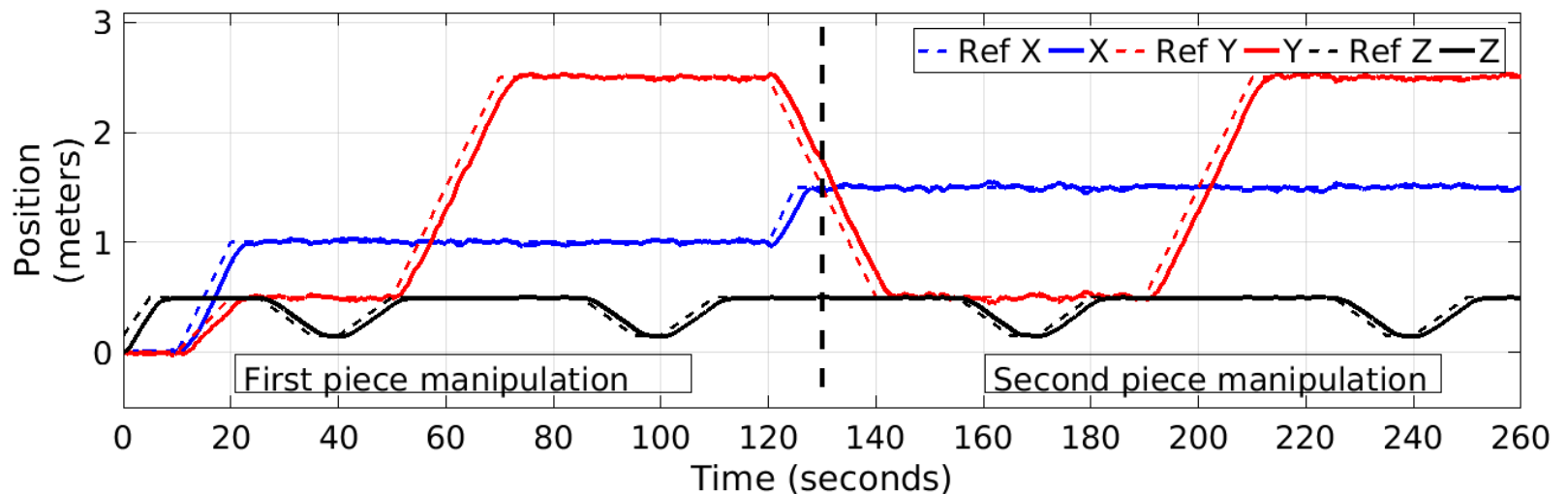
Results



NMPC
CARLA-tunned



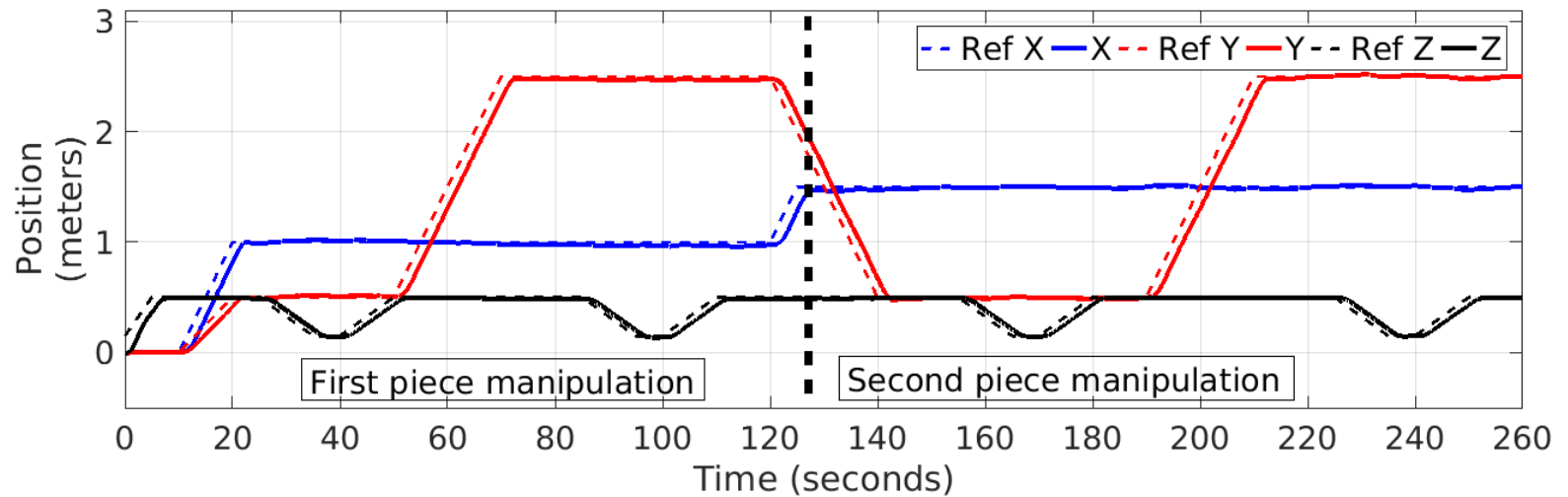
Classic PID



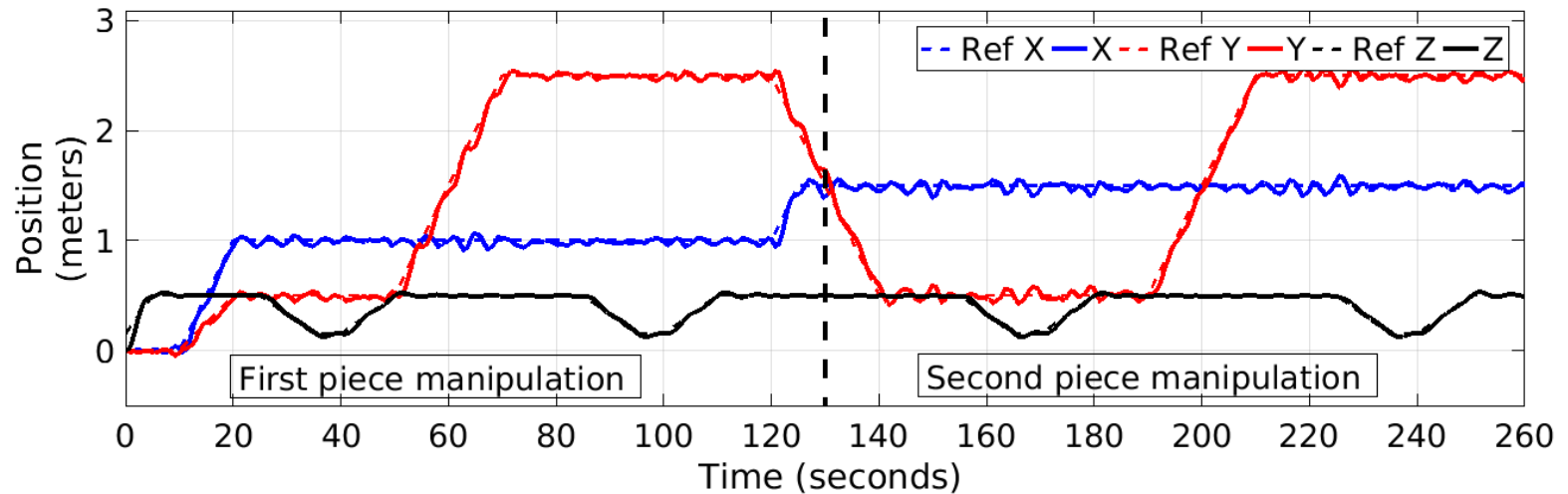
Results



NMPC
CARLA-
tuned



Nonlinear PID
learning-
based



Load Transportation



Video

Thank you!



- Questions?

References

(for presentation only)



Khalifa, Ahmed, et al. "Adaptive intelligent controller design for a new quadrotor manipulation system." Systems, Man, and Cybernetics (SMC), 2013 IEEE International Conference on. IEEE, 2013.

Goodarzi, Farhad A., Daewon Lee, and Taeyoung Lee. "Geometric stabilization of a quadrotor UAV with a payload connected by flexible cable." American Control Conference (ACC), 2014. IEEE, 2014.

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